THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 1

Chapter 5: **Describing Motion Around Us** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A runner completes one full lap of a 400 m circular track. Her displacement at the end is: 1

(a) 400 m **(b)** 0 **(c)** 200 m **(d)** 800 m

2. The slope of a velocity-time graph gives: 1

(a) speed **(b)** force **(c)** distance **(d)** acceleration

3. The equation v squared = u squared + 2as relates: 1

(a) work and time **(b)** velocity, acceleration and displacement
(c) velocity and time only **(d)** force and mass

4. The distance travelled by a body is always: 1

(a) negative
(b) less than displacement
(c) greater than or equal to the magnitude of displacement
(d) equal to velocity

5. The area under a velocity-time graph gives: 1

(a) speed **(b)** time **(c)** acceleration **(d)** displacement

6. Acceleration due to gravity near the Earth's surface is about: 1

(a) 100 m/s squared **(b)** 10 m/s **(c)** 1 m/s squared **(d)** 10 m/s squared

7. Displacement is a: 1

(a) unit **(b)** vector quantity **(c)** scalar quantity **(d)** constant

8. A straight line parallel to the time axis on a distance-time graph indicates that the body is: 1

(a) accelerating **(b)** falling
(c) moving with uniform speed **(d)** at rest

9. A ball falls from rest for 2 s. Its velocity (g = 10 m/s squared) is: 1

(a) 10 m/s **(b)** 20 m/s **(c)** 5 m/s **(d)** 40 m/s

10. Speed is a: 1

(a) force **(b)** vector quantity **(c)** scalar quantity **(d)** unit of length

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Differentiate between distance and displacement. 2

12. Define acceleration. Give its SI unit. 2

13. Why is uniform circular motion accelerated even though the speed is constant? 2

14. Differentiate between speed and velocity. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A boy walks 3 m east and then 4 m north. Find the distance and the magnitude of displacement. 3

16. From a velocity-time graph of a body moving at 6 m/s for 10 s find the distance covered. 3

17. The odometer of a car reads 2000 km at the start and 2050 km after 1 hour. Find the average speed and say what the speedometer shows. 3

OR

A car starts from rest with an acceleration of 3 m/s squared. Find its velocity after 4 s and the distance covered. 3

18. A train accelerates from 10 m/s to 30 m/s in 10 s. Find the acceleration and the distance travelled. 3

OR

A body is thrown upwards with a velocity of 20 m/s. Find the maximum height and the time to reach it (g = 10 m/s squared). 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Derive the three equations of motion for a body moving with uniform acceleration using the velocity-time graph or algebra. 5

OR

Explain uniform circular motion with the tangent velocity direction, the relation v = 2 pi r/T, and calculate the speed of a body at the tip of a 0.5 m radius wheel with period 2 s (pi = 3.14). 5

20. A car starting from rest acquires a speed of 30 m/s in 6 s and then moves with this speed for 10 s, after which it brakes uniformly to rest in 5 s. Find the acceleration in each stage and the total distance. 5

OR

Two trains A and B of speeds 72 km/h and 54 km/h are moving on parallel tracks in the same direction. Find the speed of A relative to B and the time A takes to overtake B if A is 100 m behind and 150 m long and B is 150 m long. 5

END OF QUESTION PAPER — Total: 40 Marks

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